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Porject 11 : Two robots, narrow aisles, one sparse reward



Shared aisles: Two robots fetch requested shelves through narrow aisles and can block each other.



Sparse reward: Only a delivery pays +1. A random policy delivers 0 shelves in 500 steps.



Partial view: Each robot sees a 3×3 patch. The 2-shelf request queue is hidden except for shelves inside that patch.

`RWARE tiny-2ag-v2 · 2 robots · 32 shelves · 10×11 grid · 500-step episodes`

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METHOD

IPPO with parameter sharing, task hints and optional shaping

01
IPPO:

Both robots share one 2×128 tanh actor-critic and learn with PPO from local observations.

02
Task hints (+16 features, 71 → 87 dims):

stage, next BFS step, path length, target vectors, cell-ahead-blocked.

03
Shaping (ablation):

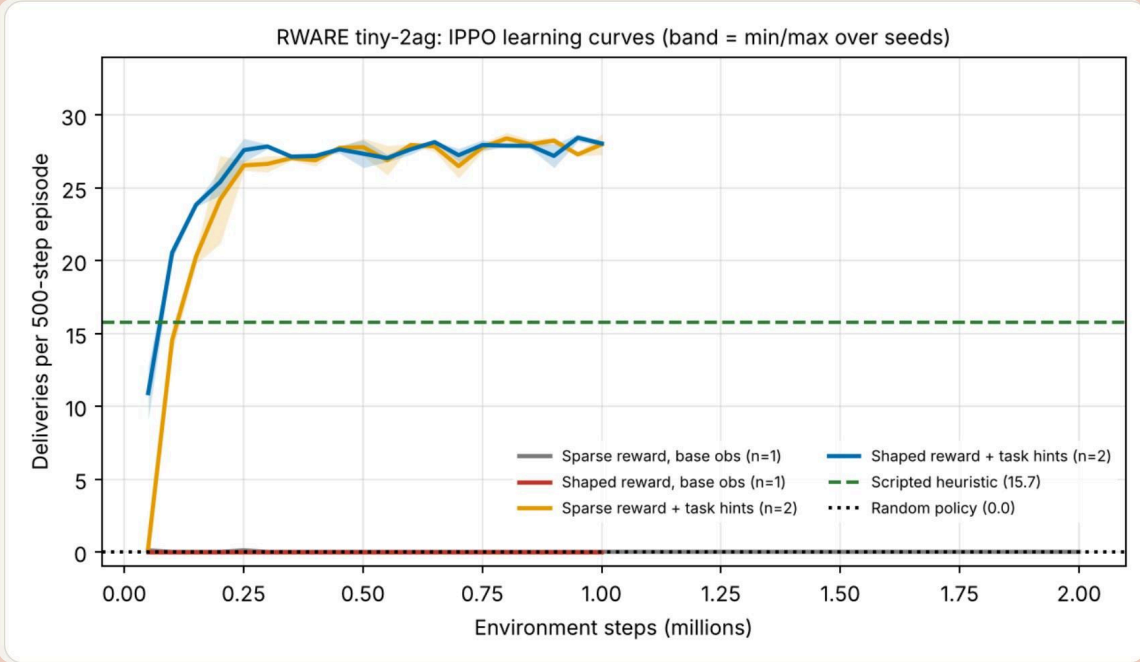
Potential-based, Φ per stage (shelf, goal, empty slot). It leaves the optimal policy unchanged.

`Reward +1 per delivery, evaluated on sparse reward only · PPO clip 0.2 · GAE λ 0.95 · γ 0.99 · 8 envs × 128 steps · time limit bootstrapped`

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RESULTS

With task hints, IPPO reaches ~24 deliveries in 150–250k steps and plateaus near 27



26.9 IPPO + hints (sparse 27.1, 26.8 · shaped 26.4, 27.3)

15.7 Scripted heuristic (episode std 10.0)

0.0 Random policy; IPPO on base observations (n=1 each, 1–2M steps)

Mean over 50 evaluation episodes per run, sampled actions, sparse reward. Two seeds for hint variants.

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LIMITATIONS

What these experiments do not show

Hints do the heavy lifting:

Base 3×3 observations delivered 0 shelves in 1–2M steps. The hints use BFS over the global grid, so the 15.7 baseline is not a like-for-like comparison.

Shaping speeds early learning, not the final score:

At 50k steps it scores ~11 against ~0, but both end at 26.9. A larger layout may differ.

Deadlocks remain:

Some episodes score 0 when robots block each other. Greedy evaluation scores lower (23–26.6) than sampled actions.

Small and simulated:

Two robots, one layout, 2 seeds.

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CONTEXT

Throughput is the reward; people are the context



Workers:

Faster fulfilment can displace jobs or raise the pace expected of human pickers.



Safety:

The reward ignores nearby people. Real robots need hard collision limits, not only learned avoidance.



Central control:

The hints assume one system knows every request, which is a single point of failure and of data collection.

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